

# Using Enterprise Architecture Knowledge Graphs as Digital Shadows for Enterprise Application Landscapes – The EAGLE Research Project

Zhuoxun Zheng<sup>1</sup> · Syed Juned Ali<sup>1</sup> · Philipp-Lorenz Glaser<sup>1</sup> · Tobias Jordan<sup>1</sup> · Sebastian Zib<sup>1</sup> · Andjela Djelic<sup>1</sup> · Thomas Hofleithner<sup>2</sup> · Franz Novak<sup>2</sup> · Emanuel Sallinger<sup>3</sup> · Dominik Bork<sup>1</sup>

the date of receipt and acceptance should be inserted later

**Abstract** Enterprise Architecture Management (EAM) increasingly supports decision-oriented tasks such as dependency analysis, impact assessment, cloud migration planning, and transformation roadmap design. However, relevant architectural knowledge is often distributed across heterogeneous sources, tools, documents, and stakeholders. The EAGLE project investigates how EAM can be improved through Knowledge Graphs (KGs) and AI methods, especially by constructing an Enterprise Architecture KG (EAKG) as a semantic foundation for analysis, reasoning, and conversational access. Using cloud migration planning as a running use case, EAGLE combines schema-guided graph construction, ontology-based task semantics, graph- and rule-based analysis, LLM-supported interaction, and human-in-the-loop validation. This report presents the project motivation, solution approach, early lessons learned, and open challenges in balancing AI-supported automation with expert control.

**Keywords** Enterprise Architecture · Knowledge Graph · Cloud Migration

## 1 EAGLE: Application Scenario and Requirements for AI-Supported EAM

EAM has long supported the documentation and governance of IT landscapes [7, 4]. With the growing complexity of digital transformation, it is increasingly expected

to provide actionable support for understanding, evaluating, and steering architectural change [11, 8]. Typical activities include application portfolio assessment, dependency analysis, impact analysis, cloud migration planning, technology rationalization, risk assessment, and transformation roadmap design [5, 9, 18]. In practice, however, most EAM activities still rely on manually maintained models, fragmented data sources, expert judgment, and tool-specific representations. As a result, enterprise architects often need to conduct a series of workshops and interviews with different stakeholder groups, including business managers, technical experts, application owners, vendor support teams, and operations staff, before they can obtain sufficient information for concrete EAM activities [19, 8]. This communication-intensive process is time-consuming and often costly, especially when architectural knowledge is distributed across people, tools, spreadsheets, models, service management systems, and informal documentation.

The EAGLE project, short for Enterprise Architecture Knowledge Graph (KG) for Learning and Exploration, addresses this gap by investigating how KGs and AI methods can reduce the manual effort required to collect, integrate, query, and analyze enterprise architecture knowledge. The project is carried out as an industry-academia collaboration between TU Wien and DIRIGENTA IT Solutions GmbH. Its central idea is to construct an Enterprise Architecture KG (EAKG) that represents heterogeneous EA-relevant information in a semantically connected form and makes this information available for analysis, prediction, and natural-language interaction [21, 6]. Rather than treating EA models as isolated diagrams or documentation artifacts, EAGLE aims to organize enterprise architecture as an interconnected and transparent knowledge base that can be

---

Corresponding author: Z. Zheng

E-mail: zhuoxun.zheng@tuwien.ac.at

1. Business Information Group, TU Wien

Erzherzog-Johann-Platz 1, 1040 Wien, Austria

2. Dirigenta IT Solutions GmbH

Bossigasse 67, 1130 Wien, Austria

3. Database and Artificial Intelligence Group, TU Wien

Favoritenstrasse 9-11, 1040 Wien, Austria

queried, extended, and used as a foundation for decision support.

The project is organized around three main research and development directions. First, EAGLE investigates the flexible construction of EAKGs from heterogeneous sources, including structured and semi-structured artifacts such as CSV and Excel files, conceptual models, and natural-language documents. This requires methods for extracting candidate schemas, entities, and relations, and mapping them to an evolving EA schema, and resolving ambiguities that arise when different sources use different names or levels of abstraction for the same architectural objects. Second, EAGLE explores EAKG-based analysis methods for architectural decision support. These include dependency analysis, impact analysis, and predictive or scenario-oriented reasoning over architectural transformations. Third, the project develops conversational access to EA knowledge, enabling users to ask questions in natural language and receive answers that are not only technically correct but also explainable with respect to the facts and relationships stored in the EAKG.

A central running use case in EAGLE is environment transformation planning, with cloud migration as a prominent example. In such scenarios, organizations need to assess how applications, data stores, platforms, and infrastructure components can be transformed from an AS-IS environment to a TARGET environment without disrupting business operations [9, 23]. This framing covers migrations to the cloud, migrations back to local data centres driven by requirements such as data sovereignty, and migrations between providers. The use case illustrates the need for integrated EA knowledge: migration decisions depend not only on technical properties of individual systems, but also on dependencies, business criticality, operational constraints, regulatory requirements, costs, performance expectations, and sustainability-related indicators [23]. By representing these factors and their relationships in an EAKG, EAGLE aims to support more evidence-based analyses, such as identifying affected business processes, estimating the impact of migration scenarios, and exploring feasible transformation roadmaps.

From these observations, EAGLE derives several requirements for AI-supported EAM. First, the EAKG must support semantic integration across heterogeneous EA data sources and tools, so that architectural knowledge can be connected beyond individual documents, models, or repositories. Second, it must remain adaptable, since enterprise architectures and their underlying schemas evolve with new technologies, organizational changes, governance requirements, and transformation projects [23]. Third, the EAKG must enable decision-

oriented analysis rather than only data lookup, including dependency reasoning, impact assessment, and scenario exploration. Fourth, the resulting knowledge must be accessible to different stakeholder groups through understandable explanations and natural-language interaction. Finally, because EA decisions involve business responsibility, compliance constraints, and expert judgment, EAGLE follows a human-in-the-loop approach in which AI methods support, structure, and accelerate EAM work without replacing domain experts. Together, these requirements are motivated by the environment transformation use case, but they also reflect broader EAM tasks in which heterogeneous architectural knowledge must be integrated, queried, analyzed, and made accessible to different stakeholders.

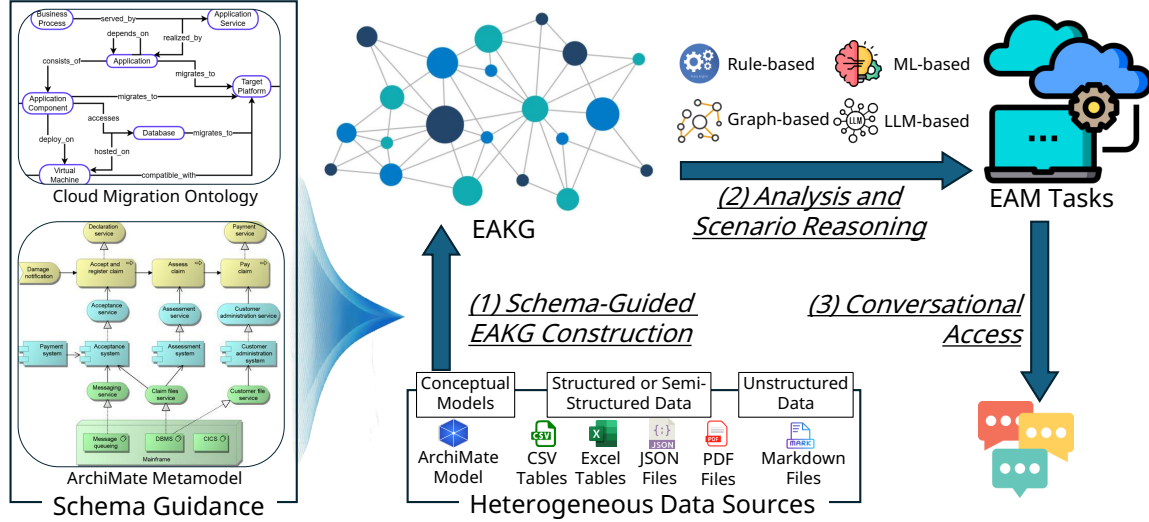
## 2 Solution Approach: From Heterogeneous EA Data to Decision Support

The EAGLE solution is centered on the EAKG [21, 6], which serves as the semantic integration and reasoning layer for AI-supported EAM. The pipeline, depicted in Fig. 1, begins with constructing connected architectural knowledge from heterogeneous EA-relevant sources (Section 2.1). The guiding schema provides a shared vocabulary for organizing extracted concepts and relations, enabling the resulting EAKG to support later querying, rule-based reasoning, and consistency checking. On this basis, the EAKG supports decision-oriented analyses such as dependency analysis, impact assessment, and scenario reasoning for architectural transformations (Section 2.2). Finally, conversational access makes these graph-based insights available to different stakeholder groups through natural-language queries and explainable answers (Section 2.3).

### 2.1 Schema-Guided EAKG Construction from Heterogeneous Sources

*Heterogeneous Data Sources.* EAGLE starts from the observation that EA-relevant knowledge is distributed across sources that differ not only in format, but also in the type of architectural knowledge they provide. Typical source data include the following:

- *Conceptual models*, such as ArchiMate diagrams [10, 12], describing the structural backbone of an EA, for example, by relating business capabilities and processes to applications and technology components.
- *Structured or semi-structured data*, such as spreadsheets or exports from management tools, often maintain application portfolios and inventories that pro-



**Fig. 1** EAGLE pipeline for EAM: (1) heterogeneous enterprise architecture data sources are integrated into an EAKG under schema guidance; (2) the resulting EAKG supports analysis and scenario reasoning for EAM tasks; (3) conversational access enables stakeholders to query, interpret, and explain graph-based results<sup>1</sup>.

vide detailed operational insights into EAM entities—including applications, databases, environments, and their mutual dependencies [2, 13].

- *Unstructured data*, found in documents like PDFs, governance policies, service-level agreements, vendor descriptions, and interview records, can capture constraints, responsibilities, rationales, and project-specific knowledge that are not explicitly represented in formal EA models [3, 15].

*Schema Guidance.* Because downstream EAM analyses require consistent concepts and relation types, EAGLE uses schema knowledge as a semantic anchor for EAKG construction. Established EA modeling concepts, in particular the ArchiMate metamodel [10], provide a useful starting point for representing business, application, and technology layers and their relationships. However, EAGLE also requires task-specific semantics that are not fully covered by general EA modeling languages. For this reason, we collaborate with DIRIGENTA to develop an ontology for cloud migration that captures concepts such as migration-relevant assets, dependency types, transformation options, operational constraints, risks, and decision criteria. Together, the ArchiMate-based schema elements and the migration-oriented ontology guide the extraction and organization of architectural knowledge from heterogeneous sources. They also provide the shared vocabulary needed for later querying, rule-based reasoning, impact analysis, and evidence-based decision support.

*Construction Process.* Based on the heterogeneous inputs and the guiding schema, EAGLE constructs the

EAKG through an iterative process of extraction, alignment, and validation. For each source, AI-supported methods first identify task-relevant architectural knowledge, such as candidate entities, attributes, relations, identifiers, constraints, and evidence fragments. The extracted knowledge is then aligned with the current EAKG schema: matching concepts and relation types are reused where possible, while missing or project-specific concepts can trigger controlled schema adaptation. The resulting nodes, edges, and properties are added to the graph, along with provenance information that links them back to their source artifacts, where possible. Since enterprise sources often contain incomplete records, implicit assumptions, and ambiguous terminology, domain experts remain involved in validating critical extraction, mapping, and adaptation decisions before the EAKG is used for downstream analysis.

## 2.2 Analysis and Scenario Reasoning over the EAKG

Once constructed, the EAKG provides the basis for decision-oriented analysis beyond simple data lookup. EAGLE investigates several complementary forms of analysis over this graph as follows.

- **Rule-based analysis.** Rules derived from the EA schema, the cloud-migration ontology, and domain expertise can be used to check constraints and derive explicit consequences of architectural situations. For example, such rules may identify migration ordering constraints, regulatory restrictions, or consistency

<sup>1</sup> The open-source icons in Fig. 1 are adapted from Flaticon (<https://www.flaticon.com>), while the ArchiMate metamodel is from Wikipedia.

violations between applications, infrastructure, and operational requirements [6].

- **Graph-based analysis.** Since the EAKG represents EAM-related units as connected entities, graph algorithms can support dependency navigation, impact propagation, clustering, and centrality-based analyses [20,6]. This enables identifying tightly coupled systems, affected components, critical hubs, and candidate groups for migration planning.
- **Machine Learning (ML)-based prediction.** Where sufficient historical, operational, or labelled data is available, ML models may be trained on EAKG-derived features to estimate quantities relevant for EAM decisions. Examples include migration effort, operational risk, expected downtime, cost effects, or selected KPI and sustainability-related indicators [22].
- **Large language model (LLM)-supported analysis.** LLMs can support the interaction with the EAKG by interpreting user questions, formulating graph queries, summarizing retrieved evidence, and suggesting possible implications in natural language [1,15]. In this role, the LLM does not replace the graph-based evidence layer, but helps stakeholders access and interpret the information represented in the EAKG.

In concrete EAM tasks, these mechanisms can be combined in different ways. In the cloud migration use case, rule-based analyses can use the EA schema and migration ontology to identify migration-relevant units with criticality, for example, by tracing how core business capabilities and processes are supported by applications, databases, platforms, and infrastructure components [6]. Graph-based analyses can complement this by examining the structure of the EAKG, using dependency patterns, connectivity, or clustering to highlight structurally central units, tightly coupled components, and potential migration groups [20,6]. LLM-supported and ML-based methods can further help analyze EAKG edges and source evidence to identify or classify candidate dependency relations between migratable units [14,17]. Based on this combined evidence, EAGLE can support enterprise architects in exploring migration alternatives, identifying sequencing constraints, and preparing wave suggestions for expert review.

### 2.3 Conversational Access to EA Knowledge

The third part of the EAGLE solution approach concerns conversational access to the EAKG. The motivation is not only to explain graph-based results, but also to make EA knowledge accessible to stakeholders who may not

be familiar with the underlying schema, graph query languages, or EAM tools. Since architectural decisions often involve business managers, application owners, technical experts, and operations staff [16], EAGLE investigates a natural-language interface that supports exploratory interaction with the EAKG. Users should be able to ask questions, refine them through follow-up interactions, and access relevant architectural evidence without having to formulate graph queries directly.

This interaction can be understood as a RAG-style process over structured EA knowledge. Instead of retrieving only text passages, the system retrieves graph facts, constraints, analysis results, and provenance from the EAKG and uses them as grounding context for LLM-supported answers. The goal is to provide responses that are understandable for different stakeholder groups while remaining connected to the underlying architectural evidence represented in the graph.

### 3 Initial Implementation and Lessons Learned

The EAGLE project has been activate since July 2025 as a close collaboration between TU Wien and DIRIGENTA. In the initial phase, we began translating the project vision into prototypes for EAKG construction and EAM support focused on cloud migration. These early activities have provided lessons on how heterogeneous enterprise data, task-specific schema guidance, and expert validation should be organized in practice.

*EAKG construction requires semantic data preparation and expert validation.* An early lesson from EAKG construction is that enterprise data cannot be directly transformed into a reliable graph without semantic preparation and validation. Even seemingly simple structured sources may contain inconsistencies that affect graph quality. For example, two CSV files may be linked by explicit IDs while containing inconsistent name fields for the same entities, making it unclear whether the issue is a data error, a synonym, or an entity mismatch. Such cases cannot be resolved reliably from the data alone and require data cleaning and expert confirmation before corresponding graph nodes and links can be trusted. We also observed that schema and ontology guidance is essential for interpreting the roles of columns and relationships in a task-relevant way. However, this mapping cannot be delegated entirely to LLMs, since many distinctions depend on domain-specific EAM and migration knowledge that is not available from general common sense alone. At the same time, purely data-driven learning is difficult because realistic enterprise datasets are limited, customer-specific, and heterogeneous in their modeling conventions. Our current approach, therefore,

combines LLM-generated candidates with human validation: AI methods reduce the manual effort by proposing possible mappings, identifiers, and semantic interpretations, while domain experts confirm or correct the decisions that determine the final EAKG structure.

*Decision support requires task-oriented semantics.* A second lesson is that an instance-level EAKG is not sufficient for decision-oriented analysis unless it is interpreted through task-specific schema or ontology semantics. In the cloud migration use case, for example, it is not enough to know that two instances are connected. The analysis also needs to interpret the schema-level pattern behind the connection. A relation between an application and a virtual machine may indicate an infrastructure dependency, whereas a relation between two applications may indicate a runtime or data dependency. Such patterns determine which reasoning rules can be applied, what information still needs to be checked, and whether the connection may create a sequencing constraint for migration planning.

The cloud-migration-oriented ontology, therefore, plays a role beyond providing a common vocabulary. It allows us to define task-specific rules that extract decision-relevant indicators from the instance graph, such as dependency categories, dependency depth, affected migration units, criticality evidence, or potential wave constraints. Even when final decision rules are not yet fully formalized, the ontology can structure the reasoning task by specifying which graph patterns and evidence are relevant for comparing alternatives or detecting constraints. This experience has shifted our implementation focus from building only a general-purpose EA graph toward constructing ontology-guided graph views that expose the information needed by human experts for concrete decisions.

## 4 Open Challenges

A major challenge in EAGLE is the highly task- and customer-specific nature of EAM. In practice, EAM rarely follows a fully standardized procedure. The relevant concepts, data sources, quality criteria, decision rules, and acceptable trade-offs often depend on the current task, the available documentation, and the requirements or preferences of the involved stakeholders. This is one major reason why traditional EAM work remains strongly manual: enterprise architects not only apply predefined rules, but also continuously interpret incomplete information, negotiate assumptions, and adapt their analysis to the specific customer context.

This creates a central challenge for EAGLE: how far can AI-supported EAM be automated without losing

expert control and contextual correctness? Too little automation would leave the communication-intensive and repetitive parts of EAM unchanged. Too much automation, however, may produce misleading decision-support suggestions that ignore customer-specific priorities. The goal is therefore not full automation, but calibrated automation: the system should automate repetitive extraction, retrieval, checking, and summarization tasks where possible, while keeping experts in control of semantic decisions, task definitions, and high-impact recommendations.

One possible direction is to make customer- and project-specific preferences explicit during the EAM process. For example, an organization may prefer conservative migration planning and prioritize business continuity over cost reduction. Such preferences can be captured as reusable constraints, rules, and annotations and stored together with the EAKG or the task-specific project configuration. Over time, these preferences could guide later extraction, schema mapping, dependency interpretation, and decision-support suggestions for the same customer. In this way, EAGLE could reduce the need for repeated manual clarification while still preserving the contextual knowledge that makes EAM decisions meaningful.

## 5 Outlook and Next Steps

EAGLE investigates how KGs and AI methods can support EAM by turning heterogeneous architectural information into a connected and task-oriented knowledge base. The project shows that AI-supported EAM should not be understood as a replacement for enterprise architects, but rather as a way to reduce repetitive information gathering, make architectural evidence explicit, and support more transparent decision-making. Early implementation experience, especially in the cloud migration use case, highlights the importance of schema guidance, ontology-based task semantics, and human-in-the-loop validation.

In the next phase, we will further develop the EAKG construction pipeline, refine the cloud-migration-oriented ontology, and extend the prototype for dependency analysis, migration wave planning, and conversational access to EA knowledge. We will also focus on evaluation, including the definition of competency questions, expert validation procedures, and benchmark-style datasets for representative enterprise sources. More broadly, EAGLE aims to contribute practical insights into how AI, Knowledge Graphs, and expert interaction can be combined to make EAM more evidence-based, adaptable, and accessible.

**Acknowledgements** This study was partially funded by the FFG-funded project EAGLE – Enterprise Architecture Knowledge Graph for Learning and Exploration (Grant number: FO999925702).

**Conflict of interest** The authors declare that they have no conflict of interest.

## References

1. Ali, S.J., Zheng, Z., Bork, D.: Shortcut or understanding? diagnosing llm type prediction in conceptual models. In: International Conference on Advanced Information Systems Engineering, pp. 37–55 (2026)
2. Chen, W., Hess, C., Langermeier, M., Stuelpnagel, J., Diefenthaler, P.: Semantic enterprise architecture management. In: International Conference on Enterprise Information Systems, vol. 2, pp. 318–325. SCITEPRESS (2013)
3. Ehrensperger, R., Sauerwein, C., Breu, R.: Collecting and integrating unstructured information into enterprise architecture management: A systematic literature review. *ICEIS* (2) pp. 728–737 (2020)
4. Fettke, P., Di Francescomarino, C.: Business process management and artificial intelligence: Literature survey and future research. *KI-Künstliche Intelligenz* **39**(2), 67–79 (2025)
5. Franke, U., Flores Rocha, W., Johnson, P.: Enterprise architecture dependency analysis using fault trees and bayesian networks. In: IR-EE-ICS.2009: 004, pp. 209–216 (2009)
6. Glaser, P.L., Ali, S.J., Sallinger, E., Bork, D.: Model-based construction of enterprise architecture knowledge graphs. In: International Conference on Enterprise Design, Operations, and Computing, pp. 57–73. Springer (2022)
7. Gong, Y., Janssen, M.: The value of and myths about enterprise architecture. *International Journal of Information Management* **46**, 1–9 (2019)
8. Hanschke, I.: *Strategisches Management der IT-Landschaft: Ein praktischer Leitfaden für das Enterprise Architecture Management*. Carl Hanser Verlag GmbH Co KG (2023)
9. Jamshidi, P., Ahmad, A., Pahl, C.: Cloud migration research: a systematic review. *IEEE transactions on cloud computing* **1**(2), 142–157 (2013)
10. Josey, A., Lankhorst, M., Band, I., Jonkers, H., Quartel, D.: An introduction to the archimate® 3.0 specification. White Paper from The Open Group **35** (2016)
11. Lamine, E., Thabet, R., Sienou, A., Bork, D., Fontanili, F., Pingaud, H.: Bprim: An integrated framework for business process management and risk management. *Computers in Industry* **117**, 103199 (2020)
12. Lankhorst, M.M., Proper, H.A., Jonkers, H.: The anatomy of the archimate language. *International Journal of Information System Modeling and Design (IJISMD)* **1**(1), 1–32 (2010)
13. Lnenicka, M., Komarkova, J.: Developing a government enterprise architecture framework to support the requirements of big and open linked data with the use of cloud computing. *International Journal of Information Management* **46**, 124–141 (2019)
14. Melo, W.: Querying enterprise architecture repositories with natural language. In: Conference on Systems Engineering Research, pp. 107–119. Springer (2025)
15. Pagels, C., Hacks, S., Bemthuis, R.: Large language models for analyzing enterprise architecture debt in unstructured documentation. In: Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing, pp. 814–822 (2026)
16. Piccirilli, E., Di Ciccio, C., Montali, M., Peñaloza, R., Pontieri, L., Ricca, F.: Explainable knowledge-aware process intelligence: Pinpoint final project report. *KI-Künstliche Intelligenz* **39**(4), 311–316 (2025)
17. Purmani, S.S.R.: Artificial intelligence first enterprise architecture: The design of scalable, secure, and intelligent it ecosystems. *American Journal of AI Cyber Computing Management* **6**(1 (2)), 1–8 (2026)
18. Ramchand, K., Baruwal Chhetri, M., Kowalczyk, R.: Enterprise adoption of cloud computing with application portfolio profiling and application portfolio assessment. *Journal of Cloud Computing* **10**(1), 1 (2021)
19. Simon, D., Fischbach, K., Schoder, D.: Enterprise architecture management and its role in corporate strategic management. *Information Systems and e-Business Management* **12**(1), 5–42 (2014)
20. Smajevic, M., Bork, D.: Towards graph-based analysis of enterprise architecture models. In: International Conference on Conceptual Modeling, pp. 199–209. Springer (2021)
21. Smajevic, M., Hacks, S., Bork, D.: Using knowledge graphs to detect enterprise architecture smells. In: IFIP Working Conference on The Practice of Enterprise Modeling, pp. 48–63. Springer (2021)
22. Takeuchi, H., Husen, J.H., Tun, H.T., Washizaki, H., Yoshioka, N.: Enterprise architecture-based metamodel for machine learning projects and its management. *Future generation computer systems* **161**, 135–145 (2024)
23. Zhao, J.F., Zhou, J.T.: Strategies and methods for cloud migration. *international Journal of Automation and Computing* **11**(2), 143–152 (2014)